



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE RAJAMURUGAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

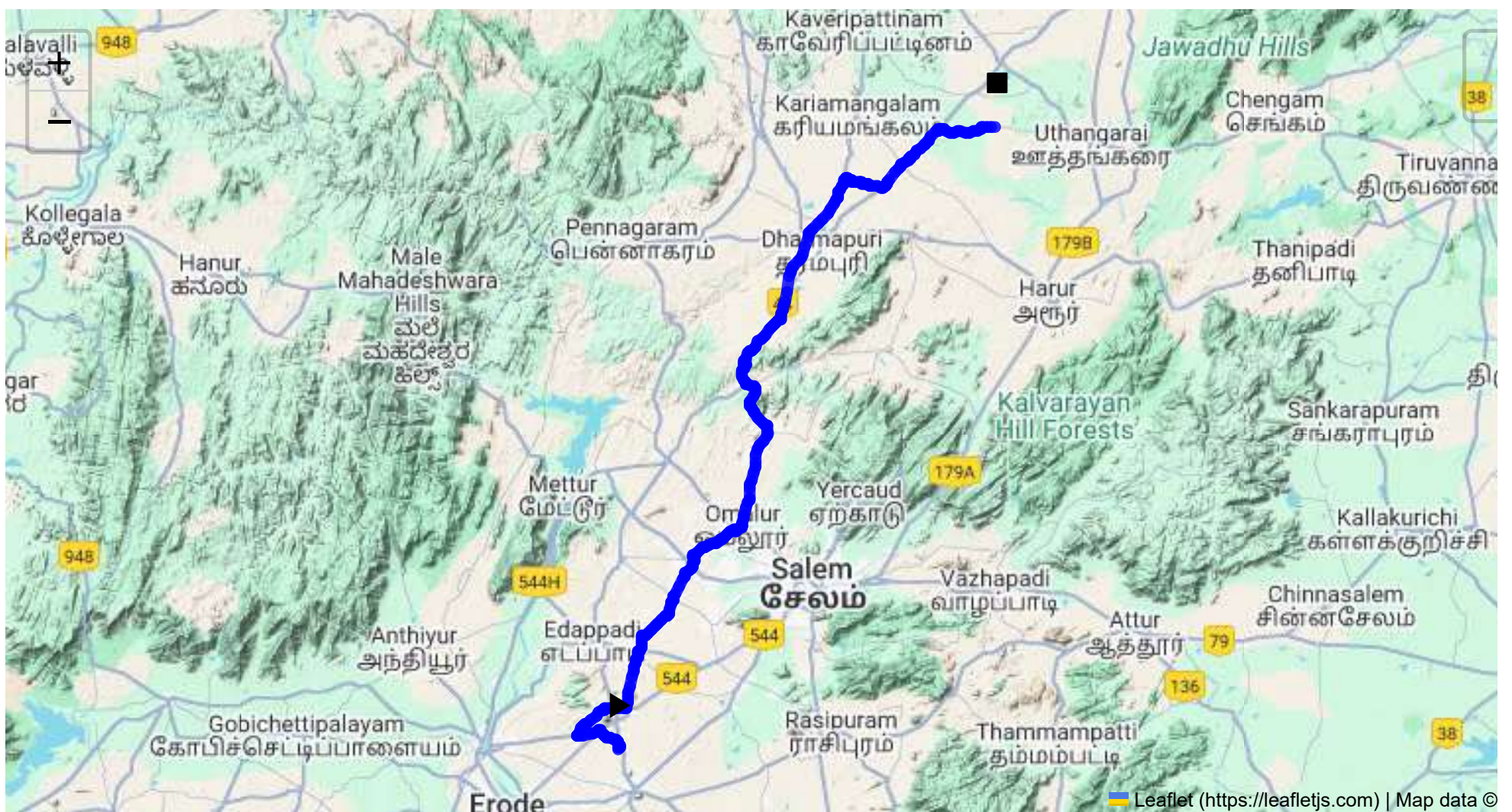
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 143.19 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.8 hours
Start: (11.4381, 77.8734)
End: (12.289012, 78.400055)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Parandapalli via various crucial points and state highways primarily follows NH544 and SH86. It traverses several small towns and rural areas, indicative of a mix of highway and local road driving conditions. The journey covers approximately 143 kilometers.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. Common weather-related hazards include:

- **Monsoon Rains (June to September):** Can cause localized flooding and reduced visibility, particularly on smaller roads.
- **Summer Heat (March to May):** Can impact vehicle performance, especially regarding tire pressure and engine temperature.
- **Fog:** Though not prevalent, morning fog during winter months can reduce visibility.

3. Traffic Patterns, Highlighting Peak Hours and Congestion-Prone Areas

- **Peak Hours:** Typically from 8:00 AM - 10:00 AM and 5:00 PM - 8:00 PM, especially near urban centers and towns.
- **Congestion Areas:** More likely around Sankari RS and Bhavani Main Road junctions due to local traffic.

4. Assessment of Road Quality and Infrastructure

- **NH544:** Generally well-maintained, suitable for heavy vehicles.
- **SH86:** Mixed conditions; paved but may have sections with potholes or narrow lanes.
- **Local Roads:** May have lower maintenance levels, leading to uneven surfaces and occasional obstructions.

5. Suggestions for Alternative Routes for Emergencies

In case of roadblocks or emergencies:

- **Alternative Route:** Consider using NH44 which runs parallel at some sections, offering a more stable pathway bypassing smaller towns.
- **Real-time Navigation Apps:** Utilize for updates on accidents or road closures and to reroute as necessary.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permissions Required:** Necessary clearances from state transport authorities.
- **Time Restrictions:** Heavy vehicles are often restricted during peak hours in urban areas.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Incidents involving overturned heavy vehicles on SH86 due to sharp turns and narrow lanes.

- Reports of chemical spillages: rare but possible, highlighting a need for strict adherence to safety protocol.

8. Environmental Considerations and Sensitive Areas

- **Proximity to Water Bodies:** Be cautious of spillages, especially near rivers and lakes in the area.
- **Habitat Conservation Zones:** Avoid pollution and adhere to speed limits to prevent disturbing wildlife habitats near forested sections.

9. Analysis of Communication Coverage, Noting Potential Dead Zones

- **Telecommunications Coverage:** Generally good in urban areas and along NH544. Potential dead zones in rural patches, particularly along SH86.
- **Recommendations:** Equip trucks with backup communication devices like satellite phones for critical updates during emergencies.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas (e.g., Sankari):** Roughly 20-30 minutes due to the proximity of emergency services.
- **Rural Areas (e.g., near Erumaipatti):** May exceed 45 minutes; alert local services promptly in case of an incident.

11. Overall Summary of Risk Assessment

This route requires careful driving due to:

- Variable road conditions, particularly on smaller roads.
- Weather-related hazards during monsoon and fog in winter.
- Peak hour congestion impacting travel time.
- Adequate preparations for emergencies, considering communication coverage and response times.

Recommendations:

- Schedule transport outside peak hours.
- Familiarize with local regulations and ensure all necessary permits are obtained.
- Constantly monitor weather forecasts.
- Utilize technology for navigation and communication to mitigate unforeseen risks effectively.

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	High	11.45615, 77.81677	15 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	Medium	11.64171, 77.95184	30 KM/Hr
16	Turn	Medium	11.64243, 77.95198	30 KM/Hr
17	Turn	Medium	11.64721, 77.95118	30 KM/Hr
18	Turn	Medium	11.67870, 77.96846	30 KM/Hr
19	Turn	High	11.70227, 77.97864	15 KM/Hr
20	Turn	Medium	11.71715, 77.99969	30 KM/Hr
21	Turn	Medium	11.71921, 78.01133	30 KM/Hr
22	Turn	Medium	11.72737, 78.02248	30 KM/Hr
23	Turn	Medium	11.73775, 78.03465	30 KM/Hr
24	Turn	High	11.73651, 78.03878	15 KM/Hr
25	Turn	Medium	11.74189, 78.04660	30 KM/Hr
26	Turn	Medium	11.74309, 78.04725	30 KM/Hr
27	Turn	Medium	11.77961, 78.05775	30 KM/Hr
28	Turn	Medium	11.86873, 78.08374	30 KM/Hr
29	Turn	Medium	11.87966, 78.08187	30 KM/Hr
30	Turn	Medium	11.92609, 78.06513	30 KM/Hr
31	Turn	Medium	11.93719, 78.05372	30 KM/Hr
32	Turn	Medium	11.95457, 78.04685	30 KM/Hr
33	Turn	Medium	11.95688, 78.04803	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
34	Turn	High	11.95828, 78.04983	15 KM/Hr
35	Turn	Medium	11.95997, 78.04946	30 KM/Hr
36	Turn	Medium	11.96335, 78.05141	30 KM/Hr
37	Turn	High	11.96408, 78.05131	15 KM/Hr
38	Turn	Medium	11.96511, 78.04966	30 KM/Hr
39	Turn	High	11.96608, 78.04919	15 KM/Hr
40	Turn	Medium	11.96819, 78.05088	30 KM/Hr
41	Turn	Medium	12.17005, 78.16792	30 KM/Hr
42	Turn	Medium	12.19919, 78.18172	30 KM/Hr
43	Turn	High	12.21894, 78.19193	15 KM/Hr
44	Turn	Medium	12.21638, 78.19617	30 KM/Hr
45	Turn	High	12.21364, 78.21156	15 KM/Hr
46	Turn	High	12.21458, 78.21208	15 KM/Hr
47	Turn	Medium	12.21314, 78.21522	30 KM/Hr
48	Turn	High	12.21236, 78.21567	15 KM/Hr
49	Turn	Medium	12.21112, 78.22080	30 KM/Hr
50	Turn	High	12.21128, 78.22322	15 KM/Hr
51	Turn	High	12.21051, 78.22351	15 KM/Hr
52	Turn	Medium	12.20519, 78.24453	30 KM/Hr
53	Turn	Medium	12.20756, 78.24733	30 KM/Hr
54	Turn	Medium	12.26786, 78.30431	30 KM/Hr
55	Turn	Medium	12.26958, 78.30708	30 KM/Hr
56	Turn	Medium	12.28427, 78.32026	30 KM/Hr
57	Turn	High	12.28657, 78.32429	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59214, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71715, 77.99969



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74189, 78.04660



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74309, 78.04725



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77961, 78.05775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86873, 78.08374



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87966, 78.08187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95457, 78.04685



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95688, 78.04803



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96335, 78.05141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96408, 78.05131



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96511, 78.04966



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17005, 78.16792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19919, 78.18172



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.21894, 78.19193



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21638, 78.19617



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21364, 78.21156



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21458, 78.21208



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21314, 78.21522



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.21236, 78.21567



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.21128, 78.22322



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21051, 78.22351



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20519, 78.24453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20756, 78.24733



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26786, 78.30431



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26958, 78.30708



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.28427, 78.32026



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.28657, 78.32429
